Marketing Analytics

Recommendation Systems and Customer Goals

By Nathan Yang

Session 12

A Returning-Customer Decision

- An mHealth user returns to log a meal.
- The system chooses among eligible categories using earlier histories.
- The objective may include familiar relevance and a stated variety goal.

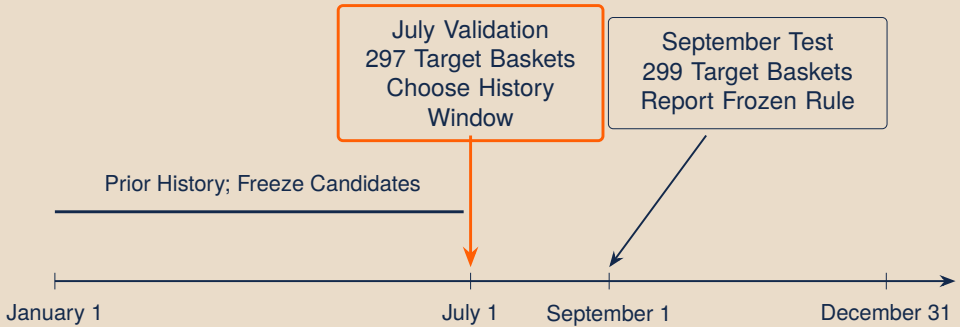
The Candidate Set

- Candidate generation determines what can be recommended.
- Scoring compares eligible items using current information.
- Ranking chooses order and applies list constraints.

Recommendation Models

- Content models compare item attributes and customer histories.
- Collaborative models learn patterns across customers and items.
- Popularity and personal-frequency rules provide useful baselines.

Histories Must Precede the Basket



All features precede each origin.
Each target is the first observed basket within that origin's month.

mHealth evaluation freezes histories and candidate sets at the forecast origin.

Recall Among Relevant Categories

Let L_i be the top-k list and B_i the categories in the next observed basket.

$$\text{Recall@k}_i = \frac{|L_i \cap B_i|}{|B_i|}$$

The denominator is the number of relevant basket categories for that user-event.

Position Matters

Discount lower-ranked relevant items when their positions affect exposure.

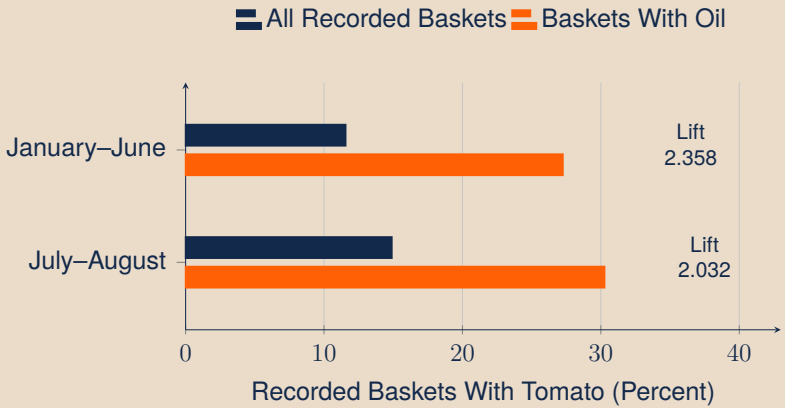
$$\mathrm{DCG@k} = \sum_{r=1}^{k} \frac{2^{\mathrm{rel}_r} - 1}{\log_2(r+1)}$$

NDCG divides by the best attainable DCG for the same relevance labels.

Basket Associations

- Support counts how often items occur together.
- Confidence conditions on the antecedent item.
- Lift compares confidence with the consequent’s marginal prevalence.

Lift Changes the Baseline Comparison



mHealth basket rules use distinct baskets and future validation.

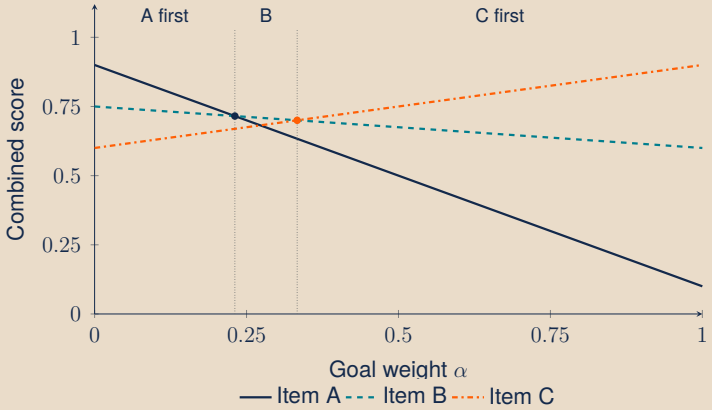
Relevance and a Customer Goal

Combine relevance s_j with a goal score g_j using an explicit weight.

$$v_j(\alpha) = (1 - \alpha)s_j + \alpha g_j$$

The scores need a comparable scale before the weight has a clear interpretation.

Small Weight Changes Can Change the First Item



Synthetic item scores. Relevance and goal scores stay fixed.

A Three-Item Basket Check

The list contains A, B, and C. The basket contains A and C.

```
from math import log2
ranked = ['A', 'B', 'C']
relevant = {'A', 'C'}
recall = len(set(ranked) & relevant) / len(relevant)
dcg = sum((x in relevant) / log2(r+2)
          for r,x in enumerate(ranked))
ideal = 1 + 1/log2(3)
assert recall == 1.0
```

Recall is 1.0, while NDCG is about 0.920 because B occupies the second position.

LLM Explanations

- Explain a recommendation using verified item attributes.
- Check whether stated reasons are factually supported.
- Evaluate explanation effects separately from ranking quality.

Offline and Online Outcomes

- A hit in a historical basket measures predictive overlap.
- A recommendation changes exposure and may change behavior.
- A randomized policy comparison measures the intervention outcome.

Coding Exercise: Ranking With a Goal

- Compare popularity and personal-frequency baselines.
- Tune the goal weight on validation baskets.
- Evaluate the fixed choice on later baskets.
- Assess one generated explanation against item attributes.

Customer Control of Recommendations

- A customer can accept a suggested goal or reject it.
- Factual explanations should distinguish evidence from an inferred preference.
- Evaluate whether the interface respects a revised goal.

Recommendation Discussion

- Which relevant item was excluded before scoring?
- Whose goal determines the ranking weight?
- What evidence would show improved customer experience?